

Real-Time Multi-Task Skin Lesion Classification and Segmentation Using Swin Transformer

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Abstract—Early identification of suspicious skin lesions can support timely clinical assessment, but automated analysis remains challenging because dermoscopic images often contain visually similar lesion categories and highly variable lesion characteristics. The objective of this study is to develop and evaluate a unified multi-task deep learning framework that performs seven-class skin lesion classification and pixel-level lesion segmentation within a single model while maintaining practical inference speed. To achieve this objective, a Swin Transformer encoder is used to extract hierarchical image features, which are shared by a classification branch and a U-Net-based segmentation decoder. Experiments were conducted on the HAM10000 dataset. Weighted cross-entropy loss and weighted random sampling were applied to reduce the effect of class imbalance, while the segmentation branch was optimized using Binary Cross-Entropy and Dice losses. On the held-out test set, the proposed model achieved 63.54% accuracy, a Macro-F1 score of 64.32%, and a ROC-AUC of 93.97% for classification. For segmentation, it obtained a Dice score of 93.98% and an IoU of 89.36%. The complete multi-task model achieved 72.58 FPS with an average latency of 13.77 ms per image under the experimental benchmark. Compared with ResNet18 + U-Net and EfficientNet-B0 + U-Net, the proposed framework achieved the highest accuracy, ROC-AUC, Dice, and IoU, while EfficientNet-B0 + U-Net achieved the highest Macro-F1 score. These findings show that the proposed framework provides strong joint classification and segmentation performance while retaining real-time-capable inference under the tested configuration.

Index Terms—Skin Cancer, Dermoscopic Images, Vision Transformer, Swin Transformer, Multi-Task Learning, Lesion Classification, Lesion Segmentation, Deep Learning, HAM10000, Real-Time Healthcare Applications.

I. INTRODUCTION

Automated analysis of dermoscopic images remains challenging due to high inter-class visual similarity, intra-class variations in texture, color, and lesion boundaries, and severe class imbalance in widely used datasets [8], [23]. These factors make consistent lesion assessment difficult and motivate computer-aided systems that provide objective support for skin lesion analysis [23].

Deep learning has become widely used for automated skin lesion analysis. Convolutional Neural Networks (CNNs), often initialized through transfer learning, have demonstrated strong performance in dermoscopic image classification [4]. For lesion localization, U-Net and its variants, including modified U-Net and multiscale attention U-Net, are widely used because their encoder-decoder structures with skip connections preserve spatial information and generate pixel-level masks [17], [18]. However, CNN-based feature extraction mainly relies on

local receptive fields, which can limit the modeling of long-range spatial dependencies and global morphological relationships. Such relationships are important because lesion shape, border irregularity, and overall structure can help distinguish visually similar categories [1].

Vision Transformer (ViT)-based models address this limitation by using self-attention to model relationships between image patches [1], [2]. Several studies have applied Transformer-based architectures to skin lesion classification with promising results [13], [14]. However, standard global self-attention has computational complexity that grows quadratically with the number of image patches, increasing computational requirements at higher image resolutions. This is an important limitation for applications requiring rapid inference [14].

Another limitation is the separate treatment of classification and segmentation [4], [17]. Classification predicts the lesion category, whereas segmentation identifies its spatial extent. A limited number of studies have investigated multi-task frameworks using shared representations or hybrid CNN-Transformer architectures [3]. However, inference speed and per-image latency are not always evaluated comprehensively. Therefore, an effective multi-task framework should jointly consider classification performance, segmentation quality, and computational efficiency.

To address these gaps, we propose a unified multi-task framework for seven-class skin lesion classification and pixel-level segmentation using a Swin Transformer encoder with a U-Net-based decoder. The Swin Transformer employs shifted-window self-attention to enable cross-window information exchange while providing hierarchical feature representations with lower computational complexity than global self-attention. The extracted features are shared by a classification head with global average pooling and a segmentation decoder with upsampling layers, allowing both tasks to be performed within a single network. The framework is evaluated on the HAM10000 dataset and compared with ResNet18 + U-Net and EfficientNet-B0 + U-Net baselines.

The primary contributions of this work are as follows:

- 1) A Swin Transformer + U-Net-based multi-task architecture is developed for joint seven-class lesion classification and pixel-level segmentation using shared hierarchical feature representations.
- 2) Class imbalance is addressed using weighted cross-entropy and weighted random sampling, while the segmentation branch uses Binary Cross-Entropy and Dice

loss. The proposed model achieves a Dice score of 93.98% and an IoU of 89.36%.

- 3) The framework is evaluated in terms of classification, segmentation, model complexity, and inference speed. Under the experimental benchmarking setup, the proposed model achieves 72.58 FPS with an average latency of approximately 13.77 ms per image, while obtaining the highest accuracy, ROC-AUC, Dice, and IoU among the evaluated models.

II. LITERATURE REVIEW

Skin lesion analysis has been widely studied using deep learning architectures. This section reviews CNN-based classifiers, segmentation networks, and Transformer-based or hybrid models, focusing on their technical strengths, limitations, and practical applicability.

CNN-Based Classification and Segmentation

Convolutional Neural Networks (CNNs) have been widely used for skin lesion classification. Early studies showed that CNNs could learn discriminative features directly from dermoscopic images and outperform traditional handcrafted feature-based methods [22], [26]. Transfer learning with ImageNet-pretrained models such as ResNet and VGG became common for addressing limited medical data and achieving reasonable performance on benchmark datasets [12], [20]. Custom CNN architectures have also reported promising melanoma detection performance [24], [25]. However, CNN-based models can still suffer from inter-class confusion, particularly between melanoma and benign nevi, where both local appearance and global morphological context are important [21]. Ensemble CNN methods can improve accuracy but often increase inference time and computational cost [7]. In addition, most CNN-based classifiers focus only on categorization and do not provide lesion localization [9], [12].

For lesion localization, U-Net and its variants are widely used because their encoder-decoder structures with skip connections preserve spatial information and generate pixel-level masks. Recent approaches, including dual-encoder designs and improved skip connections, have been proposed to enhance boundary detection and multi-scale feature extraction [5], [19]. Although these modifications can improve segmentation accuracy, they often increase model complexity. Moreover, these networks mainly rely on local convolutional operations, making long-range spatial dependency modeling more difficult, particularly for irregular lesion boundaries [5].

Vision Transformers and Hybrid Architectures

Vision Transformers (ViTs) provide an alternative to CNNs by using self-attention to capture relationships across image patches. ViT-based models have achieved competitive or superior skin lesion classification performance compared with CNN baselines, particularly when global structural context is important [15], [16]. Multi-resolution Transformer architectures have also been explored for histopathological and

dermoscopic image analysis to capture features across different scales [10].

Hybrid CNN-ViT architectures combine CNN-based local feature extraction with Transformer-based global context modeling and have shown strong performance on HAM10000 [6], [11]. Fusion approaches combining ConvNeXt with attention mechanisms have also reported promising diagnostic results [11]. However, such hybrid models can introduce architectural redundancy and additional parameters. More importantly, inference speed and per-image latency are not consistently reported in dermoscopic image studies [6], [10], limiting assessment of their suitability for real-time applications.

Multi-Task Learning and Research Gap

A limited number of studies have investigated multi-task frameworks that jointly perform classification and segmentation using shared representations. Although shared architectures can reduce redundant feature extraction, existing studies mainly emphasize predictive accuracy and Dice scores, with less attention to model size and inference speed [6], [11]. Consequently, the trade-off between computational cost and real-time capability remains underexplored. Class imbalance is another important challenge in HAM10000, where minority lesion categories may be underrepresented during training [7], [12].

In summary, CNN-based, Transformer-based, and hybrid methods each provide specific advantages for skin lesion analysis. However, relatively few studies evaluate classification performance, segmentation quality, and real-time inference together within a unified framework. Comprehensive reporting of FPS and per-image latency also remains limited in multi-task dermoscopic analysis. This gap motivates the development of a shared multi-task framework that performs classification and segmentation while maintaining real-time inference under a defined experimental setup.

III. METHODOLOGY

A. Dataset and Preprocessing

The proposed framework was developed using the HAM10000 dataset, which contains 10,015 dermoscopic images belonging to seven skin lesion categories: Actinic Keratoses (AKIEC), Basal Cell Carcinoma (BCC), Benign Keratosis-like Lesions (BKL), Dermatofibroma (DF), Melanoma (MEL), Melanocytic Nevi (NV), and Vascular Lesions (VASC). Corresponding binary lesion masks were used to support the segmentation task.

Initially, dermoscopic images, segmentation masks, and diagnosis labels were matched using the image identifier provided in the metadata file. Samples with missing masks or labels were excluded. All images and masks were resized to 224×224 pixels and normalized before training. To improve model generalization and reduce overfitting, data augmentation techniques including horizontal flipping, vertical flipping, random rotation, scaling, and brightness-contrast adjustment were applied during training. Finally, the processed dataset was divided into training, validation, and testing subsets.

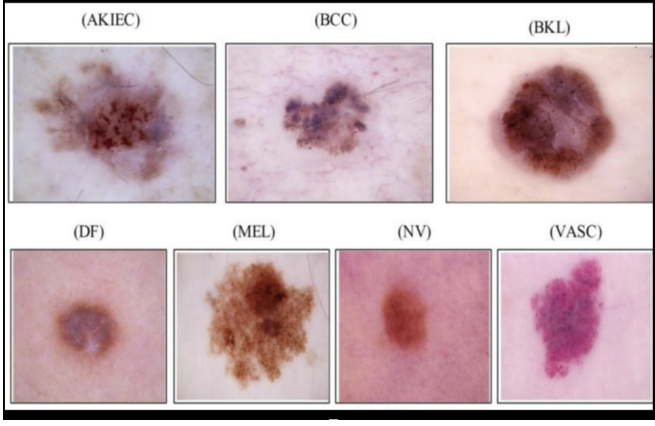


Fig. 1. Representative dermoscopic images from the HAM10000 dataset across seven lesion classes.

TABLE I
HAM10000 DATASET DISTRIBUTION

Class	Samples
AKIEC	327
BCC	514
BKL	1099
DF	115
MEL	1113
NV	6705
VASC	142
Total	10015

B. Proposed Swin Transformer-U-Net Framework

A multi-task learning framework was developed to simultaneously perform skin lesion classification and segmentation. Architecture-wise, it is made up of a Swin Transformer encoder, which is followed by two branches specific to each task. This Swin Transformer encodes hierarchical feature representations through shifted window self-attention, which allows for modeling not only local texture features but also global context.

Features encoded by the Swin Transformer are then used by the classification branch and segmentation branch. The former uses Global Average Pooling followed by a dense connection and a Softmax activation layer to make a prediction out of seven possible lesions. At the same time, the U-Net-inspired decoder up-samples feature maps and creates a lesion mask at the pixel level. By sharing feature representations between both tasks, the framework reduces redundant feature extraction and simultaneously learns diagnostic and localization information.

The classification output is defined as

$$P = \text{Softmax}(WF + b) \quad (1)$$

where F denotes the extracted feature vector, while W and b represent trainable parameters.

The segmentation output is generated as

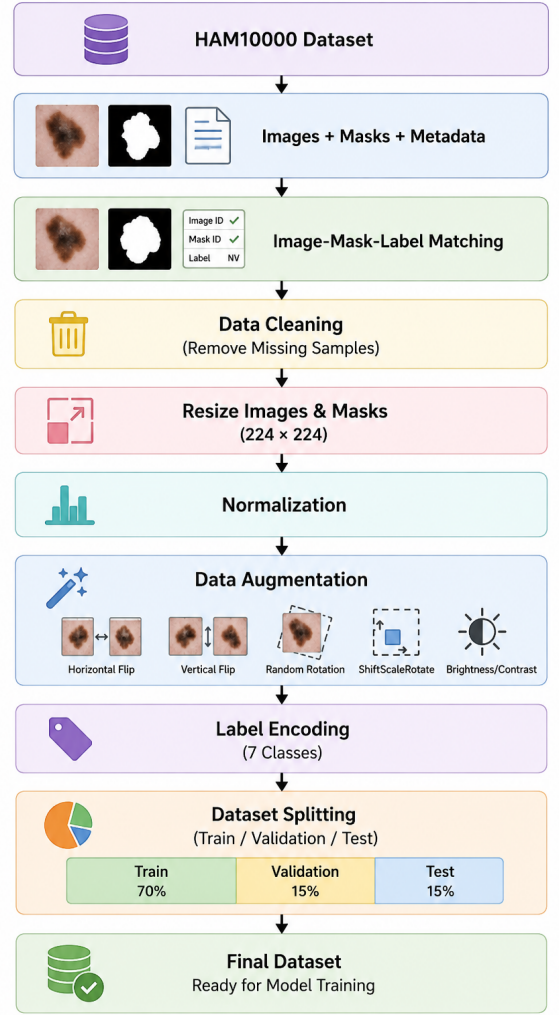


Fig. 2. Data preprocessing pipeline used for preparing dermoscopic images, segmentation masks, and diagnosis labels.

$$M = \text{Decoder}(F) \quad (2)$$

where M denotes the predicted lesion mask.

C. Loss Function

To jointly optimize classification and segmentation tasks, a multi-task loss function was employed. Due to class imbalance in the HAM10000 dataset, weighted cross-entropy loss was used for classification:

$$L_{cls} = - \sum_{i=1}^C w_i y_i \log(\hat{y}_i) \quad (3)$$

where w_i denotes the class weight, y_i is the ground-truth label, and $\hat{y}_i$ is the predicted probability.

For segmentation, Binary Cross-Entropy (BCE) loss and Dice loss were combined:

$$L_{seg} = L_{BCE} + L_{Dice} \quad (4)$$

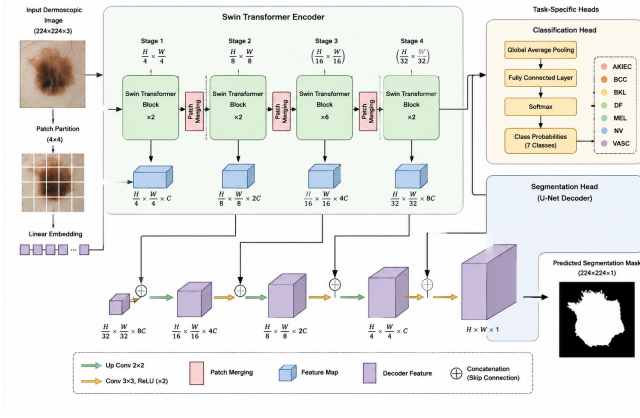


Fig. 3. Architecture of the proposed Swin Transformer and U-Net based multi-task learning framework.

TABLE II
TRAINING HYPERPARAMETERS

Hyperparameter	Value
Input Image Size	224×224
Batch Size	8
Epochs	10
Optimizer	AdamW
Initial Learning Rate	1×10^{-4}
Classification Loss	Weighted Cross-Entropy
Segmentation Loss	BCE + Dice Loss
Sampler	Weighted Random Sampler
Number of Classes	7

The overall loss function is expressed as

$$L_{total} = 0.5L_{cls} + 0.5L_{seg} \quad (5)$$

which assigns equal importance to both tasks during training.

D. Training Configuration and Evaluation Metrics

For comparative evaluation, two CNN-based multi-task baseline models were implemented: ResNet18 + U-Net and EfficientNet-B0 + U-Net. These baseline models were trained and evaluated using the same dataset split, preprocessing steps, loss functions, training configuration, and evaluation metrics as the proposed Swin Transformer + U-Net framework. This ensured a consistent comparison among the evaluated models.

The proposed model was implemented using PyTorch and optimized using the AdamW optimizer with an initial learning rate of 1×10^{-4} . Training was performed using a batch size of 8 for 10 epochs. A Weighted Random Sampler was employed during training to reduce the effect of class imbalance in the HAM10000 dataset. The hyperparameters used in this study are summarized in Table II.

The classification performance was evaluated on the held-out test set using Accuracy, Macro-F1 Score, and ROC-AUC. Segmentation performance was measured using the Dice Similarity Coefficient and Intersection over Union (IoU). In addition, inference latency, Frames Per Second (FPS), model

size, and trainable parameter count were reported to assess real-time inference capability and model complexity.

IV. RESULTS AND DISCUSSION

This section presents the results of the proposed Swin Transformer + U-Net framework on HAM10000. The evaluation covers classification, segmentation, class-wise behavior, inference speed, and model complexity. ResNet18 + U-Net and EfficientNet-B0 + U-Net were evaluated under the same protocol to compare predictive performance and complexity-efficiency trade-offs.

A. Classification Performance

Table III summarizes the classification results. The proposed model achieved the highest accuracy of 0.6354 and ROC-AUC of 0.9397. Compared with the strongest baseline for each metric, accuracy improved by 0.0306 over ResNet18 + U-Net and ROC-AUC by 0.0025 over EfficientNet-B0 + U-Net.

EfficientNet-B0 + U-Net achieved the highest Macro-F1 score of 0.6613, compared with 0.6432 for the proposed model. This indicates more balanced class-wise performance for EfficientNet-B0 + U-Net, while the proposed model achieved stronger overall accuracy and ROC-AUC. However, the absolute classification accuracy of 63.54% remains moderate and requires further improvement. The severe class imbalance in HAM10000 and the visual similarity among classes such as MEL, NV, and BKL may contribute to these remaining classification errors.

B. Segmentation Performance

The proposed Swin Transformer + U-Net achieved the highest Dice coefficient of 0.9398 and IoU of 0.8936. Relative to the strongest CNN-based baselines, this corresponds to gains of 0.0040 in Dice and 0.0062 in IoU.

These results show that the framework can generate accurate lesion masks while simultaneously performing classification. The gains may result from hierarchical contextual representations learned by the Swin Transformer, which complement local spatial information reconstructed by the U-Net decoder. Although moderate, the improvements are consistent across both segmentation metrics.

C. Confusion Matrix Analysis

The normalized confusion matrix in Figure 4 shows the highest diagonal recognition rate for Dermatofibroma (DF) at 0.94, followed by Benign Keratosis-like Lesions (BKL) and Vascular Lesions (VASC) at 0.90. Basal Cell Carcinoma (BCC) achieved 0.86, while Actinic Keratoses (AKIEC) and Melanoma (MEL) each achieved 0.78.

Melanocytic Nevi (NV) was the most challenging class, with a diagonal value of 0.57. About 0.27 of NV samples were predicted as MEL and 0.11 as BKL, while 0.13 of MEL samples were classified as BKL. The largest errors therefore occur among NV, MEL, and BKL, which share similar pigmentation, texture, and border appearance.

These results show why overall accuracy alone is insufficient. Although the framework achieved the highest overall

TABLE III
OVERALL PERFORMANCE COMPARISON OF EVALUATED MODELS

Model	Accuracy	Macro-F1	ROC-AUC	Dice	IoU	FPS	Latency (ms)	Params (M)	Size (MB)
ResNet18 + U-Net	0.6048	0.5921	0.9326	0.9353	0.8874	147.41	6.78	19.18	73.22
EfficientNet-B0 + U-Net	0.5988	0.6613	0.9372	0.9358	0.8874	82.42	12.13	15.08	57.70
Proposed Swin + U-Net	0.6354	0.6432	0.9397	0.9398	0.8936	72.58	13.77	37.06	142.43

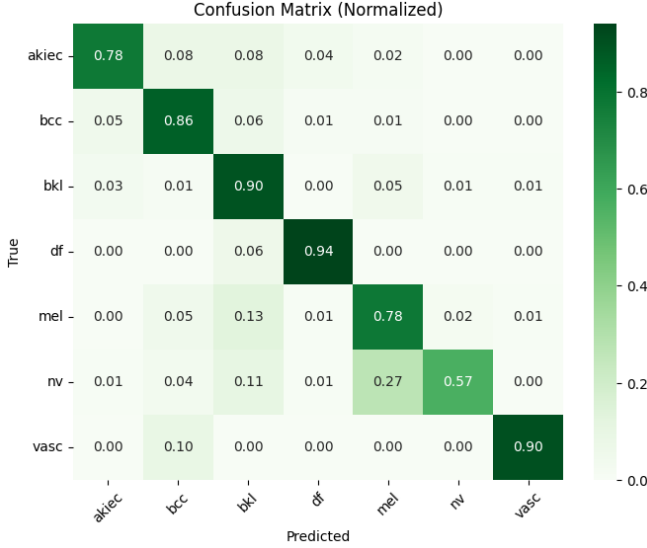


Fig. 4. Normalized confusion matrix of the proposed Swin Transformer + U-Net model.

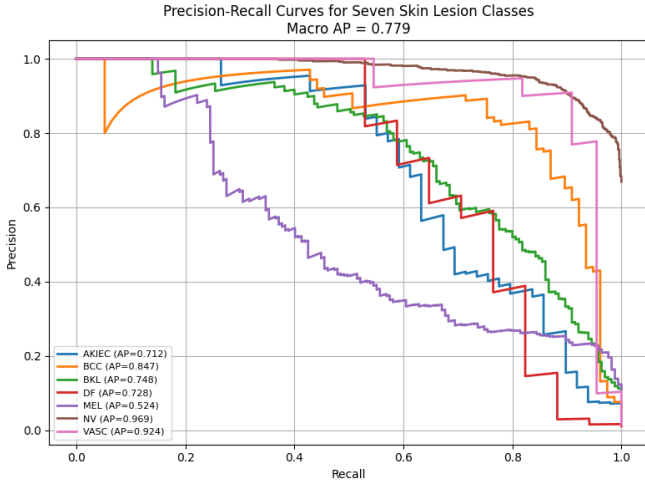


Fig. 5. Precision-Recall curves of the proposed Swin Transformer + U-Net model for seven skin lesion classes.

accuracy, its performance varies across classes, supporting class-balanced and class-wise metrics.

D. Precision-Recall Analysis

Figure 5 presents the Precision-Recall curves for all seven classes. NV achieved the highest Average Precision (AP) of 0.969, followed by VASC at 0.924 and BCC at 0.847. BKL,

DF, and AKIEC obtained AP values of 0.748, 0.728, and 0.712, respectively, while MEL achieved the lowest AP of 0.524. The overall Macro Average Precision was 0.779.

The low AP for MEL confirms that melanoma remains the most difficult class in terms of the precision-recall trade-off, consistent with its confusion with BKL and NV. Although NV achieved a high AP of 0.969, its confusion-matrix diagonal value was only 0.57. This difference reflects one-versus-rest probability ranking across thresholds versus the final multi-class prediction based on the highest class score.

These findings explain why the proposed model achieved the highest overall accuracy and ROC-AUC but not the highest Macro-F1. Improving discrimination among MEL, NV, and BKL remains important future work.

E. Real-Time Performance and Model Complexity

Inference was benchmarked on an Apple M4 system with 16 GB unified memory using the MPS backend. All models used 224×224 inputs, batch size 1, evaluation mode, and disabled gradients.

After 10 warm-up passes, 50 timed iterations were performed. Each pass executed both classification and segmentation branches; therefore, the reported latency and FPS represent complete multi-task inference.

ResNet18 + U-Net achieved 147.41 FPS at 6.78 ms, EfficientNet-B0 + U-Net achieved 82.42 FPS at 12.13 ms, and the proposed Swin Transformer + U-Net achieved 72.58 FPS at 13.77 ms.

Although slower than both CNN baselines, the proposed model still exceeds 70 FPS while performing both tasks in a single forward pass, demonstrating real-time-capable model-level inference under the tested configuration. However, preprocessing, image acquisition, interface overhead, and prospective clinical evaluation were not included.

The proposed model contains 37.06 million parameters and occupies 142.43 MB, compared with 19.18 million parameters and 73.22 MB for ResNet18 + U-Net, and 15.08 million parameters and 57.70 MB for EfficientNet-B0 + U-Net.

Overall, the proposed model improves accuracy by 0.0306, ROC-AUC by 0.0025, Dice by 0.0040, and IoU by 0.0062 over the strongest corresponding baselines, but requires greater computational resources and lower throughput. Its advantage is stronger classification and segmentation performance while retaining real-time-capable inference.

F. Overall Performance Comparison

No single model performs best across all criteria. ResNet18 + U-Net achieves the highest inference speed and lowest

latency, EfficientNet-B0 + U-Net provides the highest Macro-F1 score and lowest model complexity, and the proposed Swin Transformer + U-Net achieves the highest accuracy, ROC-AUC, Dice, and IoU.

These results address the study objectives: the framework jointly performs seven-class classification and pixel-level segmentation, achieves the strongest overall accuracy, ROC-AUC, and segmentation metrics, and reaches 72.58 FPS with 13.77 ms latency under the Apple M4 setup. Nevertheless, the moderate absolute classification accuracy indicates that further improvement is needed, particularly for challenging and visually similar lesion classes. This performance is obtained at the cost of greater computational complexity and lower throughput than the CNN-based baselines.

V. CONCLUSION

This study presented a Swin Transformer + U-Net-based multi-task framework for simultaneous skin lesion classification and segmentation. On the HAM10000 test set, the proposed model achieved 63.54% accuracy, a Macro-F1 score of 0.6432, a ROC-AUC of 0.9397, a Dice score of 0.9398, and an IoU of 0.8936. It achieved the highest accuracy, ROC-AUC, Dice, and IoU among the evaluated models, although EfficientNet-B0 + U-Net obtained a higher Macro-F1 score.

The proposed model is larger than the CNN-based baselines, with 37.06 million parameters and a size of 142.43 MB. However, it achieved 72.58 FPS with 13.77 ms average latency on an Apple M4 system using the MPS backend, demonstrating real-time-capable multi-task inference under the tested configuration.

The main limitations are the moderate classification accuracy, use of a single dataset, lack of external or multi-center validation, class imbalance, visual similarity among MEL, NV, and BKL, and model-level rather than end-to-end clinical latency evaluation. Future work will focus on larger and multi-center datasets, improved minority-class discrimination, and lighter Transformer architectures to improve classification performance, generalizability, and computational efficiency.

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